Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/15/2012
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
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City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: End date extended to 12/31/2014 per NSSC information (Ed., 9/17/13) NOTE: End date extended to 12/31/2013 per PI (Ed., 10/12/2012)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Sunderland, Peter (University of Maryland)		
Grant/Contract No.:	NNX10AD98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignition, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition (flashpoint) and extinction (fire point) can be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station and aboard parabolic aircraft. The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and record the measured surface temperature. This will allow it to use a small number of gaseous fuels, mixed with diluent, to match the following four properties of a large number of condensed fuels: heat of combustion, heat of gasification, surface temperature, and laminar smoke point. The gas flow rate will be adjusted such that the heat flux to the burner surface matches the heat required to vaporize the appropriate flow rate of liquid or solid fuel. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The flight objectives for BRE are: 1. Burn gaseous fuels in BRE to simulate the burning rates of condensed fuels in a quiescent microgravity environment. Instantaneous flame heat flux will be measured and used to control the rate of fuel supply. Fuels to be simulated include paper, plastic, and alcohols. 2. Burn gaseous fuels in BRE to estimate the ignition and extinction behavior of these condensed fuels in microgravity. 3. Use BRE to estimate the effects of burning at enhanced oxygen conditions in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	Michael Bustamante received his MS degree, his thesis entitled: EXPERIMENTAL INVESTIGATION OF LIQUID AND GAS FUELED FLAMES TOWARDS THE DEVELOPMENT OF A BURNING RATE EMULATOR (BRE) FOR MICROGRAVITY APPLICATIONS, is available from the University of Maryland. A paper and presentation was made at the Dayton Spring Technical Meeting of the Central States Section of the Combustion Institute April 22–24, 2012 entitled: Laminar Burning on Flat Wicks at Various Orientations by M.J. Bustamante, K.T. Dotson, Y. Zhang, P.B. Sunderland, J.G. Quintiere. The paper contains experimental and theoretical analysis on the burning of flat plates in natural convection at orientation angles of +90 to - 90 (with 0 being vertical). It also shows how a burner can emulate the burning of methanol and ethanol saturated on the plates. In addition aircraft flight and drop tower experiments explored the burning of these alcohols on flat plates. Quasi-steady flames were indicated in some cases, suggesting the possibility of steady burning in micro-gravity. A 50 mm diameter burner was designed and constructed to investigated the emulation of pool fires. Two heat flux meters recorded the flame heat flux at two point on the surface. By matching the flow rate of the burner fuel to the burning rate of the pool fire, and using a fuel and diluent mixture to best match the heat of combustion and smoke point, the heat of gasification was duplicated for methanol, heptane and PMMA. So three condensed phase fuels were emulated to demonstrate the validity of the burner concept. In August a SRD report was presented for review, and the Committee gave positive feedback in supporting the project's continuation in the ACME space station experimental program.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Dissertations and Theses	Bustamente MJ. "Burning Rate Emulator." M.S. Thesis, University of Maryland, Department of Fire Protection Engineering, August 2012. , Aug-2012
Papers from Meeting Proceedings	Bustamante MJ, Dotson KT, Zhang Y, Sunderland PB, Quintiere JG. "Laminar Burning on Flat Wicks at Various Orientations." Spring Technical Meeting of the Central States Section of the Combustion Institute, Dayton, OH, April 22–24, 2012. Spring Technical Meeting of the Central States Section of the Combustion Institute, Dayton, OH, April 22–24, 2012. , Apr-2012